

ADA Architecture 2.0.1

Consolidated Control, Priority, and Fault Architecture

Operational Logic, Tank Priority, Thermal Model, Fault Supervision, and
Operator Controls

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Location	Kathmandu, Nepal
Date	August 4, 2026
Document	Architecture Specification, Version 2.0.1

This document is the complete, self-contained description of the Automatic Dispatch Architecture (ADA) as designed: the thermal control law for T5, heater staging, sensor filtering, tank scoring and selection, PO tank designation, the oil-distribution scheme, dispense duration estimation, fault/E-Stop supervision, and operator override and information controls.

It describes the *architecture* only – physical I/O tags, ladder addresses, and PLC implementation detail are intentionally out of scope and covered separately.

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Executive Summary

ADA controls T5's heating, decides which of the three T10 melt tanks receives priority, keeps the designated PO tank heated, supervises the whole system for faults, runs the oil dispensing line, and gives the operator a defined set of manual controls that sit alongside – never in place of – the automatic logic. Every decision the system makes ultimately reduces to one of five things:

- **How full is T5, and what should its target temperature and active heater count be right now?** (Sections 2–4)
- **Which water-line process – T5 trim-heat, PO tank heat demand, or T10 melt – gets served, and if more than one T10 tank wants attention, which one?** (Sections 8–9)
- **Is it safe to keep running, and if not, why did it stop?** (Section 12)
- **How does oil actually get to where it's needed?** – answered completely independently of the three points above. (Section 11)
- **How does an operator pause, override, or manually drive the system without ever bypassing the safety chain?** (Section 13)

The oil line and the water line do not share decision logic. The Need Array, tank scoring, and PO tank designation govern water only – P1, P2, and P3 are all water-line concerns. Oil distribution runs on its own trigger (F2) and its own flowmeter-confirmed completion signal, with no coupling to any of the water-line mechanisms. The two systems share only the plant-wide System-OK condition, the same as every other output, plus one dedicated operator-facing disable of their own (Section 13).

Every load-bearing calculation in this architecture fits comfortably in 16-bit arithmetic – nothing here requires the wide accumulators an energy-summation approach would need.

1 The Four Schedulers

Everything ADA does is classified into exactly one of four functional schedulers. This is a conceptual grouping, not a physical or programmatic boundary – it exists to make clear *which kind of job* each piece of logic is doing.

Table 1: The four schedulers and what each is responsible for.

Scheduler	Responsible for
Operational Scheduler	Reading sensors and driving outputs. Solar routing, heater output state, branch valve state, X/Y/Z destination valve routing, and running the dispense pump against the flowmeter all live here – this scheduler acts on decisions made elsewhere, it does not make them. When Automation Mode is not selected, the three CP speed outputs can instead be driven directly from a dedicated set of panel inputs, subject to the same E-Stop gating as the automatic path (Section 13).
Priority Scheduler	Deciding who gets served, across two independent domains. On the water side: the Need Array and its trigger-count-to-state mapping (Section 8), tank scoring and the incumbent/challenger switching rule (Section 9), and PO tank designation (Section 10). On the oil side, entirely separately: the FIFO queue across X/Y/Z (Section 11), gated only by the plant-wide System-OK condition, with no coupling to the water-side logic. Water-side assignment can be held off entirely on operator command without stopping the system (Section 13).
Mathematical Core	Computing the numbers the other three schedulers consume. The level division register (Section 2), $T_{\text{target}}(d)$ (Section 3), heater-count lookup and both anti-chatter latches (Section 4), sensor averaging (Section 5), T_f (Section 6), the P2 heat-demand function (Section 7), and the oil line’s pump runtime estimate (Section 11.6) are all produced here. Nothing in this scheduler drives an output or arbitrates contention directly.
Fault Scheduler	Keeping the system safe and stopping it when it isn’t – for the water line only. Physical and Virtual E-Stop, the system-enabled condition, and Trend Threshold Supervision all live here (Section 12). Tunability within Trend Threshold Supervision is deliberately asymmetric – see Section 12 for why. The oil line has no fault path of its own by design (Section 11.5); it is subject only to the shared System-OK condition and the operator’s own oil-line disable, like every other gated output.

1.1 Process Summary: Level, Temperature, and Heater Count

The most frequently-run loop in the architecture is the one that keeps T5's temperature and heater staging aligned with how full the tank currently is. In brief: the Operational Scheduler reads the raw level and temperature every scan; the Mathematical Core turns the raw level into a filtered division count and looks up the corresponding target temperature and nominal heater count; the two anti-chatter latches then decide whether it is actually time to act on that lookup; and only then does the Operational Scheduler update the live heater outputs.

2 The Level Division Register

T5 is 2000 mm tall. Level is tracked in **5 mm divisions** rather than raw millimetres, giving a convenient 0–400 register range for every lookup function that follows:

$$d = \frac{\text{height}(\text{mm})}{5}, \quad d \in [0, 400], \quad d=0 \text{ empty}, \quad d=400 \text{ full} \quad (1)$$

Mass follows directly from the height-to-mass constant established for T5 (Section 14):

$$\text{mass}(\text{kg}) = \text{height}(\text{mm}) \times 2.4353 = d \times 12.1765 \quad (2)$$

$d = 200$ is 50% full; $d = 360$ is 90% full. These two points recur throughout the architecture – as the knots of the thermal target function, as the heater-staging range, and as the P3 (T10 melt) demand trigger.

3 $T_{\text{target}}(d)$: The Thermal Control Law

T5's target water temperature is a direct, closed-form function of how full the tank currently is – no energy summation, no dynamic power evaluation, just a level lookup:

$$T_{\text{target}}(d) = \begin{cases} 70, & d < 200 \\ 70 - \frac{3}{2560}(d - 200)^2, & 200 \leq d < 360 \\ 40, & d \geq 360 \end{cases} \quad (3)$$

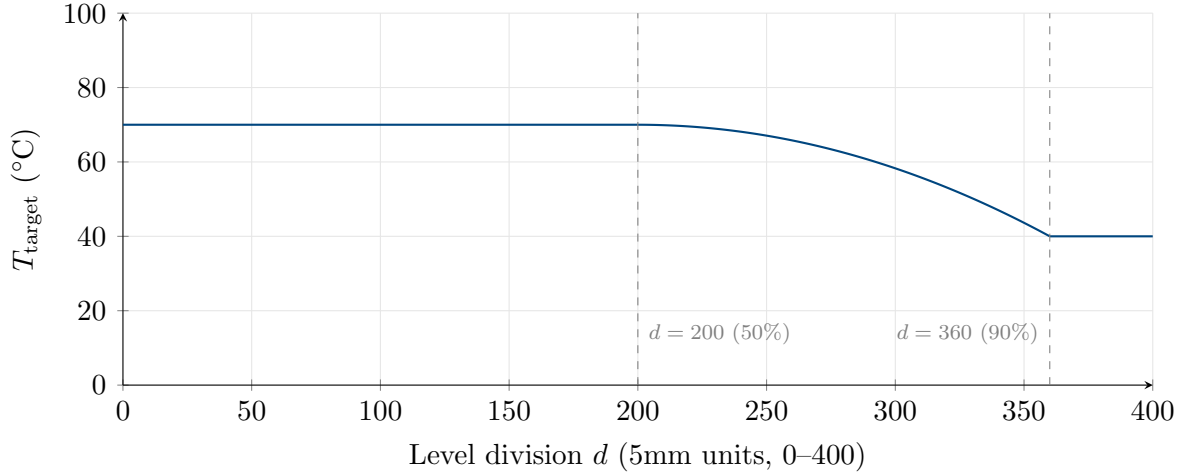


Figure 1: $T_{\text{target}}(d)$: flat 70 °C below 50% full, quadratic taper to 40 °C by 90% full, flat above.

Continuity confirmed at both knots. At $d = 200$ the quadratic term vanishes, giving exactly 70, matching the flat piece below it. At $d = 360$: $70 - \frac{3}{2560}(160)^2 = 70 - 30 = 40$, matching the flat piece above it. No discontinuity exists at either boundary.

Interpretation. Below half-full, T5 is aggressively heated toward the design ceiling of 70 °C. Between 50% and 90% full the target tapers down quadratically, so the reduction is steepest near the low end (where the most heater capacity is still engaged) and flattens out as fewer heaters remain active. Above 90% full the target drops to a minimal holding value of 40 °C – T5 needs little further heat input once it is that full.

Solar assist is gated off the same function – there is no separate temperature target for the solar loop – and, like every other output in the architecture, off the plant-wide System-OK condition (Section 12):

$$\theta_{\text{Sol}} > T_{\text{target}}(d) \Rightarrow \text{route through solar} \quad \theta_{\text{Sol}} < T_{\text{target}}(d) \Rightarrow \text{bypass solar} \quad (4)$$

Whenever solar water is hot enough to beat the current target, it is used; otherwise the loop bypasses solar and relies on the electric heaters.

4 Heater Staging: Level-Band Lookup with Dual Latches

4.1 Base Lookup

Three heater elements (Heater 1, Heater 2, Heater 3) are staged off level alone. The active band from 50% full ($d = 200$) to 90% full ($d = 360$) – a span of 160 divisions – is split into three **equal thirds** of roughly 53 divisions each, one heater dropping out per third:

Table 2: Heater count by level division (nominal, pre-latch).

Level band (d)	Heaters active
$d < 253$	3 (all heaters)
$253 \leq d < 306$	2
$306 \leq d < 360$	1
$d \geq 360$	0

Why these particular boundaries. Below 50% full, all three heaters stay on regardless – that region is absorbed into the first band. From $d = 200$ onward, the 160-division run up to $d = 360$ (90% full) is divided into three equal steps of $160/3 \approx 53.3$ divisions: $200 \rightarrow 253 \rightarrow 307 \rightarrow 360$, each step dropping exactly one heater. The table above rounds those steps to whole divisions (253, 306, 360).

4.2 Rounding Latch (Anti-Chatter, Filling Direction)

When the tank is **filling** (d increasing) and a heater-count *decrease* would nominally trigger, that decrease is held off until d reaches a slightly higher buffer point. This prevents a heater bank from rapidly cycling on/off if the level hovers exactly at a boundary.

Table 3: Rounding latch buffer points (filling direction only).

Transition	Nominal boundary	Latched release point
3 heaters $\rightarrow$ 2	$d = 253$	Hold until $d = 260$
2 heaters $\rightarrow$ 1	$d = 306$	Hold until $d = 310$
1 heater $\rightarrow$ 0	$d = 360$	Hold until $d = 370$

Open item. The latch is only specified for the filling (count-decreasing) direction. The draining direction (count-increasing transitions, e.g. $0 \rightarrow 1$ at $d = 360$) is assumed to use the plain nominal boundaries directly, with no buffer – chatter matters more when repeatedly de-energising a heater bank than when energising one. Confirm this asymmetry is intentional before this is finalised.

4.3 Comfort Latch (Anti-Chatter, Temperature-Driven)

Independent of the rounding latch above. This governs *when measured temperature actually permits cutting heat*, not how many heaters the level nominally calls for.

Table 4: Comfort latch: hold-off condition once $\theta_S \geq T_{\text{target}}(d)$.

$T_{\text{target}}(d)$ range	Hold off until
$= 70$ (i.e. $d < 200$)	Cut heaters immediately, no further hold
$65 \leq T_{\text{target}} < 70$	Wait until measured θ_S actually reaches 70°C
$T_{\text{target}} < 65$	Wait until measured θ_S reaches $T_{\text{target}} + 5$

The two latches operate on independent axes and do not conflict: the rounding latch decides how many heaters should be considered active based on level; the comfort latch decides whether the measured result is good enough to actually act on that decision. A transition can be rounding-latched (waiting for level to cross a buffer point) while simultaneously comfort-latched (waiting for temperature to catch up) – both conditions must clear before any change to the heater state.

5 Sensor Averaging

Raw analog readings are shown live everywhere they are simply displayed (e.g. HMI values), with no added lag. Anything feeding a *calculation* in this architecture instead consumes a 5-second block-averaged value, to reject transient spikes caused by agitation (oil movement disturbing a level or temperature reading).

Mechanism. A register sums the raw reading once every second. After the fifth addition, the accumulated sum is divided by 5, and that result becomes the filtered value every downstream calculation uses. The register then resets and starts summing again. The filtered value therefore refreshes once every 5 seconds and holds steady in between.

This filtering applies to every channel that feeds a calculation – ten channels in total: the four level channels (T5, TA, TB, TC) and six temperature-derived channels. The same ten channels are what Trend Threshold Supervision (Section 12) watches for suspicious rates of change, and a fixed subset of the filtered values is mirrored to a dedicated HMI trend-chart block for the operator (Section 13).

6 T_f : Dynamic Upper Thermal Boundary

To protect process safety and structural integrity, the system enforces a dynamic upper thermal limit, $T_f \leq 55^\circ\text{C}$, evaluated continuously. Unlike the lower bound (40°C , measured directly at the tank), the upper threshold is not a fixed number – it is the equilibrium temperature the system would reach if all its thermal mass mixed together, computed under a conservation-of-energy, worst-case (100% heat transfer) assumption.

6.1 Derivation

Three thermal masses are involved: the water (w), the stainless-steel structure (ss), and the palm oil being processed (p). Water and steel start at a common temperature x and cool toward the equilibrium point T_f ; the palm oil mass m_p starts at temperature y and warms toward the same T_f .

Heat lost by the water and steel cooling from x to T_f :

$$Q_{\text{lost}} = (m_w c_w + m_{ss} c_{ss})(x - T_f) \quad (5)$$

Heat gained by the palm oil mass warming from y to T_f , using palm oil's standard specific heat capacity ($c_p \approx 2000 \text{ J/kg} \cdot ^\circ\text{C}$):

$$Q_{\text{gained}} = m_p c_p (T_f - y) = 2000 m_p (T_f - y) \quad (6)$$

Setting $Q_{\text{lost}} = Q_{\text{gained}}$ and writing $K = m_w c_w + m_{ss} c_{ss} \approx 479\,060 \text{ J}/^\circ\text{C}$ for the combined, fixed thermal capacitance of the water and steel structure:

$$K(x - T_f) = 2000 m_p (T_f - y) \quad (7)$$

Solving for T_f :

$$T_f(x, y, m_p) = \frac{Kx + 2000 m_p y}{K + 2000 m_p} = \frac{479060x + 2000 m_p y}{479060 + 2000 m_p} \quad (8)$$

Dividing through by the common scalar factor of 2000 gives the simpler, equivalent form:

$$T_f(x, y, m_p) = \frac{239.53x + m_p y}{239.53 + m_p} \quad (9)$$

6.2 Interpretation and Use

- x – the current (or starting) temperature of the water and steel structure.
- y – the current (or starting) temperature of the palm oil mass m_p involved.
- T_f – the equilibrium temperature the two masses would settle at if fully mixed under worst-case heat transfer.

The system continuously evaluates $T_f(x, y, m_p)$ from live inputs. Whenever the function yields $T_f > 55^\circ\text{C}$, an interlock cycle is triggered – the architecture treats this as a genuine safety boundary, independent of and in addition to the $T_{\text{target}}(d)$ control law above. T_f is also one of a small set of values kept on permanent display for the operator via the HMI trend chart (Section 13), since it is the single number that most directly reflects how close the system is to that boundary.

7 PO Tank Demand Function (P2)

Structure only – the exact combination is set at commissioning. As with the pump runtime estimate (Section 11.6), getting this function’s internal shape right depends on real equipment behaviour, phase state transitions (solid vs. liquid palm olein), and heat exchange dynamics, not pure first-principles derivation. What is fixed here is the inputs, the state-dependent piecewise structure, and the mechanism for tuning it; the exact combining functions and thresholds are established through **real-time calibration before commissioning**.

The PO tank’s own heat demand – the estimated pump runtime D_{PO} (in seconds) required to bring the tank to its target temperature θ_{target} – is computed continuously from three live inputs: the PO tank’s own temperature, the PO tank’s mass, and the supplying water temperature.

Because palm olein undergoes a phase transition requiring latent heat of fusion, heat transfer rate changes significantly between solid and liquid phases. To account for this, D_{PO} is defined as a piecewise thermal control law conditioned on the current tank temperature θ_{tank} :

$$D_{PO}(\theta_{tank}, m_{tank}, \theta_w) = \begin{cases} C_{cal,melt} \cdot A, & \theta_{tank} < \theta_{melt} \\ C_{cal,liquid} \cdot B, & \theta_{melt} \leq \theta_{tank} < \theta_{target} \\ 0, & \theta_{tank} \geq \theta_{target} \end{cases} \quad (10)$$

where the demand functions A and B represent the uncalibrated thermal demand during the melting phase and liquid phase, respectively:

$$A = \frac{m_{tank} [c_{p,s}(\theta_{melt} - \theta_{tank}) + L_f + c_{p,l}(\theta_{target} - \theta_{melt})]}{\theta_w - \theta_{tank}} \quad (11)$$

$$B = \frac{m_{tank}(\theta_{target} - \theta_{tank})}{\theta_w - \theta_{tank}} \quad (12)$$

- θ_{tank} , m_{tank} – the PO tank’s own temperature and mass, both available as filtered (5-second averaged) values (Section 5).
- θ_w – the supplying water temperature.
- θ_{melt} , θ_{target} – the palm olein slip melting point and required operational target temperature, respectively.
- $c_{p,s}$, $c_{p,l}$, L_f – the specific heat capacity (solid), specific heat capacity (liquid), and latent heat of fusion of the palm olein batch.
- $C_{cal,melt}$, $C_{cal,liquid}$ – dedicated **calibration constants** for the melting and liquid heating regimes, respectively. Tuning these independent constants in real time during commissioning, rather than hard-coding fixed mechanical coefficients, keeps the demand check accurate as equipment and operating conditions drift over the installation’s lifetime.

P2 (Section 8) is set whenever D_{PO} crosses its operational start threshold – indicating the PO tank requires heating – and clears once D_{PO} drops below the shut-off threshold.

7.1 Demand Watchdog

D_{PO} is not allowed to sit indefinitely against a demand that never resolves. A dedicated watchdog timer runs whenever P2 is active and clears the demand state outright once it times out, rather than leaving a stale, ever-active P2 flag holding the Priority Scheduler's attention. The same clear can also be issued directly by the operator from the main HMI's reset control, independent of the watchdog timing out on its own (Section 13). Either path returns P2 to its normal, re-triggerable state – the watchdog does not raise a fault or stop the system, it simply prevents a single unresolved demand condition from permanently occupying a Need Array slot.

8 The Need Array: Three Water-Line Triggers

Scope. The Need Array is a **water-line** concept only. It governs T5's own trim-heat, the PO tank's continued heating, and T10 melt demand – it has nothing to do with, and does not trigger, oil leaving the building. Oil distribution is a separate system entirely, with its own independent flag; see Section 11.

Three triggers represent the three water-line processes that can want service. Each is referred to directly as P1, P2, or P3 – there is no separate flag letter for these three, to avoid any confusion with the oil line's own flag (F2, Section 11).

Table 5: Need Array trigger conditions.

Process	Concerns	Trigger
P1	T5 trim-heat	$\theta_S \leq 40^\circ\text{C}$
P2	PO tank heat demand	$D_{PO}(\theta_{\text{tank}}, m_{\text{tank}}, \theta_w)$ crosses threshold (Section 7)
P3	T10 melt, aggregate	T5 level < 90% ($d < 360$)

P3's trigger is an aggregate condition, not per-tank. P3 fires as a single trigger off T5's own level alone – melting demand is active whenever T5 is below 90% full, which in practice is nearly all the time, and pauses only once T5 is comfortably stocked. *Which* physical T10 tank actually receives water during an active P3 window is a separate decision, resolved by tank scoring (Section 9), not by the trigger condition itself.

8.1 Trigger Count to System State

$$\text{count}(P1, P2, P3) = 1 \rightarrow \text{Low-Speed Mode} \quad 2 \rightarrow \text{High-Speed Mode} \quad 3 \rightarrow \text{Full-Speed Mode} \quad (13)$$

With exactly three possible triggers, no four-way tie-break is ever needed at this level. Contention *within* P3 – which physical T10 tank fills that one slot – is the switching-rule question addressed below. The resulting speed mode is always visible on the HMI by name (Low/High/Full-Speed Mode), and can be temporarily held from advancing by the operator without affecting anything else in the array (Section 13).

9 Tank Scoring and Selection

9.1 The Scoring Formula

One formula is used for every T10 tank ranking decision:

$$\text{score} = 0.6\alpha + 0.4\theta \quad (14)$$

where α is the tank's level and θ is its temperature. There is no separate variant formula – this single score drives both the P3 switching rule below and the informational PO-designation evaluation (Section 10).

9.2 Selection Switching

Once a tank is locked in as the active P3 selection, it is not simply re-picked as "whichever scores highest" every cycle – that would be vulnerable to rapid switching whenever two tanks' scores are close. The switching rule instead depends on how many T10 tanks are actually eligible to compete, which in turn depends on whether a PO tank is currently designated.

9.2.1 Three Eligible Candidates (no PO tank designated)

Pick the **highest**-scoring tank as incumbent. A challenger only displaces it once the incumbent has fallen to half the challenger's score:

$$\text{score}_{\text{incumbent}} \leq 0.5 \times \text{score}_{\text{challenger}} \quad (15)$$

With three free tanks, prioritising whichever is closest to done (highest score) finishes tanks fastest and frees up capacity soonest.

9.2.2 Two Eligible Candidates (a PO tank designated)

Inverted. Pick the **lowest**-scoring tank as incumbent. The other candidate only displaces it once *that* candidate has fallen to half the incumbent's (already low) score:

$$\text{score}_{\text{other}} \leq 0.5 \times \text{score}_{\text{incumbent}} \quad (16)$$

Why the inversion, as best understood. With one tank tied up on PO duty, melt throughput is already constrained, so priority shifts from "finish the closest tank fastest" (throughput-maximising, the 3-way case) to "don't let the neediest tank fall further behind" (urgency-prioritising). This is the working rationale, not an independently-derived requirement – confirm it if a different consideration was actually intended.

Score floor, enforced. A tank score is never allowed to reach exactly 0 – the inverted rule would otherwise be unable to trigger a switch away from a zero-score incumbent ($0.5 \times 0 = 0$, and no valid score is < 0). The PLC clamps every computed score to a small positive floor value before it is used in any comparison: if $0.6\alpha + 0.4\theta$ would evaluate at or below that floor, the floor value is written in its place instead. This guarantees the inverted switching rule always has a genuine nonzero score to compare against, so an incumbent pinned at its lowest possible reading can still be displaced once a real challenger falls low enough.

Table 6: Switching rule summary.

	3 eligible (no PO)	2 eligible (PO designated)
Initial pick	highest score	lowest score
Switch condition	incumbent $\leq 0.5 \times$ challenger	other $\leq 0.5 \times$ incumbent
Optimises for	throughput (finish fastest)	urgency (don't neglect neediest)

10 PO Tank Designation

Designation of which T10 tank (TA, TB, or TC) is currently the PO tank is a **manual, operator-driven decision** – it is not automatic. The operator selects the PO tank via a virtual switch on the HMI (one of four states: none designated, or TA/TB/TC designated). Once set, that tank is removed from the melt-priority pool and becomes the tank oil is drawn from for dispensing (Section 11), which is what collapses tank scoring from three eligible candidates to two (Section 9.2). P2's own heat-demand function (Section 7) continues to run against whichever tank is currently designated, keeping it heated while it is being drawn down.

The scoring formula from Section 9 ($0.6\alpha + 0.4\theta$) is also shown on the HMI for each T10 tank as an **evaluation aid** – a live ranking to help the operator judge which tank is the best candidate to designate. It does not make the designation decision itself; the operator always confirms the final choice.

10.1 Low-Mass Advisory

If the designated PO tank's mass drops below the target mass needed to fulfil a pending oil request, the system does not intervene automatically – it simply displays a message on the HMI and continues. The operator decides what to do with that information: stop and redesignate a fuller tank (better for overall system efficiency), or proceed anyway and accept whatever shortfall results. There is no forced stop and no fault raised either way – this is deliberately left as an operator call, not a PLC decision.

11 Oil Distribution: X, Y, Z

This is a fully independent system. Oil distribution does not read, use, or in any way depend on the Need Array, tank scoring, or PO tank designation – it never has. It runs on its own trigger, its own queueing rule, and its own completion signal. Its shared dependencies with the rest of ADA are the overall System-OK condition (Section 12) and a dedicated operator-facing disable of its own (Section 13): if the whole system is halted, or the operator has switched oil dispensing off, oil dispensing stops – not because the oil line’s own decision-making needs anything from the water line to function correctly.

Palm oil can be dispensed to one of three destinations, X, Y, or Z. Each destination has its own local HMI. The oil line’s own demand condition – whether any destination currently has a pending request – is referred to as **F2**, entirely distinct from, and unrelated to, the water-line’s P1/P2/P3 triggers (Section 8) despite the shared numeral.

11.1 Single-Destination, Full-Flow Dispensing

Only one destination is ever served at a time, at full flow rate. The system does not split or throttle flow between destinations – for whichever destination is active, its valve path is opened and the other two are closed off entirely, so all pump output goes to the one selected destination.

11.2 Request Queueing: FIFO

An operator at any of the three destination HMIs enters the amount of oil required there, which raises F2. Because only one destination can be served at a time, requests are served strictly in the order they were made (first-in, first-out). When the active dispense finishes, the system moves on to the next queued request; F2 clears once the queue is empty.

11.3 Request Acknowledgement and Cancellation

When a destination raises a request, the system sends a short acknowledgement back to that destination’s own HMI, so each satellite panel can confirm its request was actually registered rather than lost in transit – a plain handshake, timed out and retried on the same schedule as the rest of the satellite communication (Section 11.7). Each destination can also cancel its own pending request before it is served; a cancellation clears the computed dispense amount for that destination back to zero, so a cancelled request cannot be picked up part-way or resume with a stale figure if re-entered later.

11.4 Prime Function

Because oil can get trapped in the piping between dispenses, each destination has a **prime function**: closing off the other two paths to isolate that single destination, then letting the operator manually run the pump (via a button on that destination’s HMI) to clear the line before an actual metered dispense. This ensures the dispensed amount is accurate rather than being short-changed by oil still sitting in the pipe from a previous run.

11.5 Flowmeter-Confirmed Completion

The flowmeter, not a timer, ends a dispense. Once a destination’s line is primed – full of oil, bar the last few drops that inevitably remain in the pipe – the pump runs at full flow rate until the flowmeter confirms the requested amount has actually left. That flowmeter reading is the sole authority on completion; nothing about timing, estimated duration, or elapsed time plays any role in deciding when a dispense is done.

There is deliberately no fault path here. If a dispense does not complete as expected, nothing about the oil line escalates, alarms, or auto-stops on its own – there’s simply nothing to detect a fault against, since the flowmeter itself is the ground truth. If a request never resolves, the operator notices and issues it again. That is the entire recovery mechanism by design; see also Section 12 for how this differs from the water line’s Trend Threshold Supervision.

11.6 Pump Runtime Estimate

Structure only – the numbers inside are not invented here, and this is display/planning information only, not a completion signal. The dominant term is how palm oil’s viscosity changes with the PO tank’s temperature, which is equipment- and product-specific; getting it wrong would look authoritative while being quietly incorrect. This needs either a manufacturer viscosity curve for the oil in use, or a short empirical characterisation (dispense a known mass at 2–3 different tank temperatures, time it, fit the curve) during commissioning.

Separately from the flowmeter’s authoritative completion signal, a runtime estimate T_{runtime} is computed purely for HMI display and planning purposes (e.g. an “estimated time remaining” readout) – it never gates, stops, or flags anything:

$$T_{\text{runtime}}(m, \theta_{\text{PO}}) = C_{\text{cal,rt}} \cdot A \cdot B \quad (17)$$

where the intermediate terms and calibration factor are defined as:

$$A = \frac{m}{\text{rate}_{\text{ref}}} \quad (18a)$$

$$B = e^{k_{\text{visc}}(\theta_{\text{ref}} - \theta_{\text{PO}})} \quad (18b)$$

$$C_{\text{cal,rt}} = \frac{T_{\text{actual}}}{A \cdot B} \quad (18c)$$

Nominal Baseline (A): Calculated directly from the HMI-entered target mass m and the reference mass flow rate rate_{ref} measured at baseline temperature θ_{ref} .

Temperature-Viscosity Correction (B): Captures the dominant viscosity correction $f_{\text{visc}}(\theta_{\text{PO}})$. Colder oil increases fluid viscosity, thereby slowing flow speed ($B > 1$), while warmer oil increases flow speed ($B < 1$). The coefficient k_{visc} is fit to empirical viscosity data during commissioning.

Calibration Constant ($C_{\text{cal,rt}}$): Derived during commissioning (or updated dynamically via PLC auto-tuning) as the ratio of true measured elapsed time T_{actual} to theoretical model prediction $A \cdot B$. This constant is independent of, and never shared with, $C_{\text{cal,P2}}$ (Section 7).

11.7 Satellite Communication

Each of the three destination HMIs is a satellite station polled on a round-robin schedule, so a request or a cancellation from any one destination is picked up within one poll cycle of the others – no destination is favoured over another by the polling order itself.

12 Fault Supervision

12.1 Physical and Virtual E-Stop

Both paths do the same fundamental job – stopping the system safely – but are raised and cleared differently.

Physical E-Stop. Raised by pressing the physical E-Stop mushroom button. Reset is simply unlatching it (twisting/releasing the button) – no separate reset action is needed.

Virtual E-Stop. Raised automatically by a fault condition detected by the system itself (e.g. Trend Threshold Supervision below tripping). Reset requires the operator to press the Reset control shown on the main HMI once the system has stopped – this only clears the stop once the underlying fault condition is no longer active. The same Reset control also clears a stalled PO demand watchdog condition (Section 7.1) if one happens to be outstanding, but the two are otherwise independent: an outstanding PO demand does not itself hold the Virtual E-Stop, and vice versa.

System-enabled condition. The system is only allowed to run when all three of the following are true simultaneously: the physical E-Stop is not tripped, *and* the virtual E-Stop is not tripped, *and* Automation Mode is selected. If any one of these three conditions fails, the system is not okay to run and all process outputs are inhibited. This single condition is what the rest of the architecture’s operational flow is gated on, and it is also the condition the manual override path in Section 13 must still respect.

12.2 Trend Threshold Supervision

Water-line only. This is the only trend-based supervision in the architecture. It watches the ten filtered water-line channels from Section 5 – the four level channels and six temperature-derived channels – the oil line has no equivalent, by design (Section 11.5).

Monitors rate-of-change per channel, every cycle, against a limit. Runs on the same 5-second cadence as sensor averaging (Section 5): each channel’s current filtered value is compared against its own value from one cycle earlier, and the ratio between the two is checked in both directions – a rise beyond the high-side limit, and a fall beyond the low-side limit – from that single computed ratio.

Tunability is asymmetric, on purpose. Only the four level channels (T5, TA, TB, TC) carry operator-adjustable high- and low-side limits; the six temperature-derived channels run against a fixed, conservative limit instead. This is a deliberate consequence-weighted choice, not an oversight: a bad or fast-moving level reading feeds directly into mass, into T_f , into tank scoring, and into which tank gets priority – if a level channel is wrong, the knock-on effects are immediate and can be severe (an overfill, a dry-run, a wrong priority decision). A temperature channel drifting or spiking is comparatively low-consequence over the timescales this system operates on, so it is left on a fixed limit rather than exposed to per-site tuning. The four channels that matter most if they fail are the four channels the operator can actually retune.

Table 7: Trend Threshold Supervision states (applied independently to each of the ten channels).

State	Condition	Behaviour
0	rate below limit	nothing
1	rate at or above limit	alarm sounds, system pauses (Virtual E-Stop raised)

There is no intermediate warning tier and no escalation path – a channel is either within its limit or it has crossed it. If any one of the ten channels crosses its limit, the alarm and system pause trigger regardless of the state of the others; the channels are evaluated independently but share the same stop action.

Clearing the trip follows the standard Virtual E-Stop reset behaviour described above: pressing the HMI Reset control only clears the stop if the offending channel has since fallen back below its limit. This re-check happens automatically on every scan, so a reset attempted while the condition still holds simply re-trips – the operator cannot silently override an active threshold breach from the HMI.

The HMI identifies which channel caused the trip and displays the value that triggered it, so the operator has what they need to inspect the discrepancy before resetting. A rolling trend chart of the most safety-relevant values is kept permanently visible on the HMI regardless of trip state, as a live benchmark for the operator (Section 13).

Open item. Whether the six temperature channels eventually warrant their own adjustable limits, the way the four level channels already do, is left as a future decision rather than settled here – for now they are treated as adequately conservative on a fixed limit.

12.3 Why the Oil Line Has No Fault Path

This is a deliberate design choice, not an oversight. The oil line's completion signal (Section 11.5) is a direct physical measurement – the flowmeter either confirms the requested amount left, or it doesn't. There is no elapsed-time estimate, no tiered warning, and no auto-stop on the oil side: if a dispense doesn't resolve as expected, the operator simply notices and re-issues the request. This mirrors the same "display it, let the operator decide" philosophy used for the P2 mass-below-target case (Section 10) – the system surfaces information, but a human makes the call rather than the PLC silently intervening.

12.4 Remaining Fault Detection

Level/mass anomaly detection (mass change while no active transfer is live) and the 75 °C T10 hard-limit alarm are independent of Trend Threshold Supervision and run continuously alongside it.

12.5 Branch Valves

Branch valves throughout the system are kept normally open; energising a branch valve closes it. No further detail is needed here at the architecture level.

13 Operator Override and Manual Controls

None of this replaces the automatic logic. Every control in this section sits alongside the schedulers described above – it gives the operator a defined way to pause, redirect, or manually drive part of the system, but it never bypasses the System-OK condition (Section 12), and it never lets the PLC's own safety checks be silently skipped.

13.1 Manual Speed Override

When Automation Mode is not selected, the operator can drive the three CP speed outputs (Low/High/Full-Speed Mode, Section 8) directly from a dedicated set of panel inputs, instead of letting the Need Array's trigger count decide the speed. This manual path is subject to exactly the same Physical and Virtual E-Stop gating as the automatic path – it is a different way of choosing the speed, not a way of running the system with the safety chain relaxed. It sits alongside the automatic Need Array logic; selecting Automation Mode at any time hands control straight back to the automatic path.

13.2 Priority Pause

The operator can hold the Priority Scheduler's water-side assignment in place from the main HMI, without stopping the system and without affecting anything else. While paused, P1/P2/P3 triggers continue to be evaluated and the current speed mode is retained, but a change in trigger count will not itself move the system to a new tank or a new speed mode until the pause is lifted. This is distinct from an E-Stop: the plant keeps running and heating, only the priority *decision* is held.

13.3 Oil Line Master Disable

A dedicated master switch at the main HMI can disable oil dispensing outright, independent of the water-line's System-OK condition. With the switch engaged, no destination can be served even if a request is queued and the rest of the system is running normally – this gives the operator a way to take the oil line out of service on its own, without needing to halt water-line operation to do it.

13.4 Operator Information Displays

Two purely informational features support the controls above and the fault supervision in Section 12, neither of which alters any control decision:

- **Trend chart.** A small, fixed set of the most safety-relevant values – including T_f (Section 6), T5's level, and T_{target} (Section 3) – is continuously mirrored into a dedicated HMI chart block, refreshed on the same tick as the rest of the filtered sensor data. Keeping this on its own register block, rather than reading the live process values directly, keeps the chart's history stable even while the underlying registers are mid-calculation.
- **Input status mirror.** The full set of physical digital inputs is mirrored to the HMI for diagnostic purposes, giving the operator a direct view of raw field-input state independent of any of the system's own interpretation of those inputs.

14 Height-to-Mass Conversion

All tanks are treated as cylindrical with a flat liquid surface. Using a density of $\rho = 957.0 \text{ kg/m}^3$:

Table 8: Height-to-mass constants.

Tank	D (m)	Mass/mm (kg)	Max height (mm)
T5	1.80	2.4353	2000
T10 (TA/TB/TC)	2.32	4.0455	2200

These constants are what Section 2's level-division register and the T_f derivation in Section 6 both ultimately rest on.

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Based on the *VCH Sizing Framework, Second Edition*
DOI: 10.5281/zenodo.21009246